Premature (Preterm) Baby

Definition, causes and stage of development

What is a Premature Baby?

A premature (preterm) baby is one born before 37 completed weeks of pregnancy. Full term is 37-42 weeks.

Categories by Gestational Age

Category	Gestational age
Extremely preterm	Before 28 weeks
Very preterm	28 to under 32 weeks
Moderate to late preterm	32 to under 37 weeks

By birth weight: low birth weight is under 2,500 g, very low is under 1,500 g, and extremely low is under 1,000 g.

Causes (Why It Happens)

Group	Examples
Maternal conditions	Infections (UTI, bacterial vaginosis, chorioamnionitis), pre-eclampsia, hypertension, diabetes, anaemia or malnutrition
Uterine and placental	Cervical insufficiency, uterine abnormalities, placental abruption or placenta previa, premature rupture of membranes
Pregnancy factors	Twins or multiples, IVF, previous preterm birth, short gaps between pregnancies
Lifestyle and other risks	Smoking, alcohol, drug use, severe stress, maternal age under 17 or over 35
Medical reasons	Labour sometimes induced early to protect mother or baby; in many cases no clear cause is found

Stage of Development and Risks

Preterm babies are born with organs that have not finished maturing:

System	Problem
Lungs	Surfactant production starts around weeks 24-28, so respiratory distress syndrome is common
Brain	Immature breathing control causes apnoea; fragile vessels risk intraventricular haemorrhage
Heart	Patent ductus arteriosus may fail to close
Gut	Feeding is difficult; risk of necrotising enterocolitis
Eyes	Retinopathy of prematurity can develop
Temperature and skin	Little body fat, so hypothermia is common

System	Problem
Other	Jaundice is more frequent; weak immune system makes infections more likely

Outcome and Key Treatments

Survival and outcomes improve sharply with each week of gestation. Viability starts around 22-24 weeks with intensive care.

- Antenatal corticosteroids (betamethasone) given to the mother
- Surfactant therapy
- Caffeine citrate for apnoea
- Incubator care